

NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 2 EXAMINATION 2023-2024
EE6403 – DISTRIBUTED MULTIMEDIA SYSTEMS

April / May 2024

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. In a transform-based image compression scheme similar to the baseline JPEG, an image is partitioned into numerous 4×4 pixel blocks. Each pixel block then undergoes two-dimensional Discrete Cosine Transform (2-D DCT), element-by-element quantization, and entropy encoding. The 4×4 DCT matrix **P** and the quantization matrix **Q** are given as follows:

$$\mathbf{P} = \begin{bmatrix} 0.5000 & 0.5000 & 0.5000 & 0.5000 \\ 0.6533 & 0.2706 & -0.2706 & -0.6533 \\ 0.5000 & -0.5000 & -0.5000 & 0.5000 \\ 0.2706 & -0.6533 & 0.6533 & -0.2706 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} 10 & 20 & 30 & 40 \\ 20 & 30 & 40 & 50 \\ 30 & 40 & 50 & 60 \\ 40 & 50 & 60 & 70 \end{bmatrix}.$$

Consider a 4×4 pixel block **A** given by:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 80 & 80 & 0 \\ 0 & 80 & 80 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Note: Question No. 1 continues on page 2.

- (a) Find the 2-D DCT of $\mathbf{A}$. (6 Marks)
- (b) Find the quantized 2-D DCT coefficients of $\mathbf{A}$. Round the quantized coefficients to the nearest integer. (3 Marks)
- (c) The quantized 2-D DCT coefficients of $\mathbf{A}$ in part (b) undergo decompression to reconstruct the pixel block. Find the decompressed pixel block $\hat{\mathbf{A}}$. (7 Marks)
- (d) Calculate the total square error between the original pixel block $\mathbf{A}$ and the decompressed pixel block $\hat{\mathbf{A}}$. (3 Marks)
- (e) The two-dimensional Discrete Fourier Transform (2-D DFT) of an $N \times N$ pixel block f_{xy} is given by:

$$F_{uv} = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f_{xy} e^{-j2\pi \left(\frac{ux+vy}{N} \right)} \quad u, v = 0, \dots, N-1.$$

State clearly whether 2-D DCT or 2-D DFT is a better transform to perform image compression. Briefly justify your answer.

(6 Marks)

2. In an image compression scheme, Singular Value Decomposition (SVD) is used to compress an image block $\mathbf{B}$ given by

$$\mathbf{B} = \begin{bmatrix} 4 & 2 \\ 5 & 3 \\ 2 & 1 \end{bmatrix}.$$

The SVD of $\mathbf{B}$ is given as $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$.

- (a) Show that one of the singular values in $\mathbf{\Sigma}$ is given by $\sigma = 7.6756$. Hence, find the matrix $\mathbf{\Sigma}$. (7 Marks)
- (b) Find the matrix $\mathbf{V}$. (7 Marks)

Note: Question No. 2 continues on page 3.

- (c) The matrix $\mathbf{BB}^T$ has an eigenvector $\mathbf{u}_1$ that corresponds to the largest eigen-value, and it is given by:

$$\mathbf{u}_1 = \begin{bmatrix} 0.5821 \\ 0.7593 \\ 0.2910 \end{bmatrix}.$$

Find the reconstructed image block using rank-1 approximation.

(6 Marks)

- (d) Consider a new image block $\mathbf{C}$ given by:

$$\mathbf{C} = \begin{bmatrix} 40 & 20 & 40 & 20 \\ 50 & 30 & 50 & 30 \\ 20 & 10 & 20 & 10 \end{bmatrix}$$

Sketch the graph of total square error of the approximated image against the number of ranks used in low-rank approximation of $\mathbf{C}$.

(5 Marks)

3. (a) In an application, a user would like to compress a video using the MPEG-2 standard.

The video formats are given as follows:

Resolution of the luminance plane: 720×480 pixels

Color depth of each pixel in each luminance and chrominance plane: 8 bits/pixel

Frame rate: 30 frames per second

The chosen MPEG formats are given as follows:

Chroma subsampling format is 4:2:2

Group-of-Pictures (GOP) structure: IBBPBBPBB.

The average compression ratios for the I-frames, P-frames, and B-frames are assumed to be 10:1, m :1, and n :1, respectively, where m and n are real positive numbers. The target bitrate of the compressed video is 5×10^6 bits/s.

- (i) In Case 1, assume $m = 25$ and $n = 50$, find the average bitrate of the compressed video in bits/s. Can the obtained bitrate meet the target bitrate requirement?
- (ii) In Case 2, assume $m = 30$, find the value of n that satisfies the bitrate requirement. State any assumption(s) you make in your calculation.

Note: Question No. 3 continues on page 4.

- (iii) Suggest two changes in the MPEG formats above that can reduce the bitrate of the compressed MPEG-2 video.
(15 Marks)
- (b) With the aid of a diagram, briefly explain the operation principle of lenticular lens in glassless 3D TV display operation.
(5 Marks)
- (c) Briefly describe the flow control process in Transmission Control Protocol (TCP).
(5 Marks)
4. (a) Draw a diagram of TCP/IP protocol stack, and clearly label all the layers. State clearly the role(s) of Transport layer in the TCP/IP protocol stack.
(8 Marks)
- (b) The bitrate of a data stream is given as 6×10^6 bits/s. The data is modulated using different modulation methods. Find the symbol rate of the data stream in symbols/s for the following modulation methods:
- (i) Quadrature Phase Shift Keying (QPSK).
- (ii) 64 Quadrature Amplitude Modulation (64-QAM).
(6 Marks)
- (c) Draw a diagram of the Transformer encoder structure out of the full Transformer architecture. Clearly label all the key components.
(6 Marks)
- (d) With the aid of an equation, briefly explain how the Transformer performs self-attention.
(5 Marks)

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.